Hypothesis Testing

Complete Exam Answer Guide

9	**5**	**2**
Questions	Step Procedure	Error Types

Based strictly on: Chapter 10 — One Sample Tests of Hypothesis (Course PPT) and 18.-Hypothesis-Testing.pdf (Tushar B. Kute) + Confidence.pptx

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Q1 | 5 Marks | 2022-23 Re-exam

What is the Difference Between a T-Test and a Z-Test? When should each be used?

What is a T-Test?

The t-test (also called Student's T-Test) compares two averages (means) and tells you if they are different from each other. It also tells you how significant the differences are — in other words, it tells you whether those differences could have happened by chance. It is used to determine if the mean (average) of two groups are truly different.

> Named after William Sealy Gosset who used the pseudonym 'Student' — hence Student's T-Test. Not because it is used in college, but because of its inventor's pen name.

What is a Z-Test?

A Z-Test is used when you DO know the population's mean and standard deviation (sigma). Since sigma is known, the normal distribution applies directly and the Z statistic can be calculated. The Z-test is used for large samples (n >= 30) from a known population.

Key Differences

Aspect	T-Test	Z-Test
Population SD (sigma)	UNKNOWN — use sample SD (s)	KNOWN — sigma is given
Sample size	Typically n < 30 (small samples)	Typically n >= 30 (large samples)
Distribution used	t-distribution with (n-1) degrees of freedom	Standard Normal distribution (Z)
Test statistic formula	t = (x-bar - mu) / (s / sqrt(n))	Z = (x-bar - mu) / (sigma / sqrt(n))
When comparing groups	Comparing means of 2 groups (unknown population)	Testing one sample mean vs known population

3 Types of T-Tests

Independent Sample T-Test: You are comparing the average of two INDEPENDENT, unrelated groups — samples from two different populations. Tests whether they have a different mean. Example: comparing test scores of students from two different schools.

Paired Sample T-Test: You compare the average of two samples taken from the SAME population at different points in time. Tests means of before-and-after observations from the same target. Example: measuring patient weight before and after a treatment program.

One-Sample T-Test: Tests if the average of a SINGLE group is different from a known or hypothesized average. Example: testing whether the mean salary at one company equals the industry average.

Decision rule: If you have MORE than 2 samples, you cannot simply run multiple T-Tests — instead use ANOVA (Analysis of Variance). T-tests are only for comparing 1 or 2 means.

Q2 | 10 Marks | 2022-23 Re-exam

Write down the steps of testing of Hypothesis with examples. (Define H0, H1, test statistic, critical region, decision rule)

What is Hypothesis Testing?

Hypothesis testing is a procedure, based on sample evidence and probability theory, used to determine whether the hypothesis is a reasonable statement and should not be rejected, or is unreasonable and should be rejected. You are essentially testing whether your results are valid by figuring out the odds that your results could have happened by chance.

Key Definitions

Null Hypothesis (H0)

The null hypothesis is always presumed to be true. It always contains equality (=, >=, <=). H0 has NO burden of proof. It is the 'status quo' or accepted fact.

Example: H0: mu = 200 (production rate has

Alternate Hypothesis (H1)

H1 carries the burden of proof. It always contains strict inequality (not=, >, <). A random sample is used to reject H0 and support H1. Accepted only when H0 is rejected.

Example: H1: mu != 200 (production rate HAS changed)

Important rules: H0 and H1 are mutually exclusive and collectively exhaustive. Equality is ALWAYS part of H0 (=, >=, <=). The symbols !=, <, > are ALWAYS part of H1. If we conclude 'do not reject H0', this does NOT mean H0 is true — only that there is insufficient evidence to reject it.

The 5 Steps of Hypothesis Testing

1 **State the Null and Alternate Hypothesis**

Clearly define H0 (null) and H1 (alternate). Identify keywords in the problem and convert them to mathematical symbols. Use the keyword table to determine direction of test. Example (Jamestown Steel): H0: mu = 200 | H1: mu != 200 (keyword: 'has changed')

2 **Select the Level of Significance (alpha)**

The level of significance alpha is the probability of committing a Type I Error (rejecting H0 when it is actually true). Commonly used values: alpha = 0.05 or alpha = 0.01. This is stated in the problem. Example: alpha = 0.01 as stated in the problem.

3 **Select the Appropriate Test Statistic**

Choose between Z-test (sigma known, large sample n >= 30) or t-test (sigma unknown, small sample n < 30). Z formula: Z = (x-bar - mu0) / (sigma / sqrt(n)) t formula: t = (x-bar - mu0) / (s / sqrt(n)) with degrees of freedom = n - 1

4 **Formulate the Decision Rule (Critical Region)**

The critical region (rejection region) is the set of values of the test statistic for which H0 is rejected. Two-tailed test: Reject H0 if |Z| > Z(alpha/2) OR |t| > t(alpha/2, n-1) Right-tail test: Reject H0 if Z > Z(alpha) OR t > t(alpha, n-1) Left-tail test: Reject H0 if Z < -Z(alpha) OR t < -t(alpha, n-1) Alternatively: Reject H0 if p-value < alpha

5 **Make a Decision and Interpret the Result**

Calculate the test statistic from the sample data. Compare with the critical value or compare p-value with alpha. State the conclusion in plain language. Example: 'Because the test statistic 1.55 does not fall in the rejection region, H0 is not rejected. We conclude the production rate has not changed from 200 per week.'

Keyword Table — Setting Up H0 and H1

Keyword / Phrase	Symbol	Part of
Larger (or more) than	>	H1
Smaller (or less) than	<	H1
No more than	<=	H0
At least	>=	H0
Has increased / Has improved (test scores)	>	H1
Is there a difference? / Has changed	!=	H1
Has not changed / Remains the same	=	H0
Has improved (medication speed / time)	<	H1

Full Worked Example — Jamestown Steel Company

Problem: Jamestown Steel makes desks. Weekly production follows Normal distribution with mu = 200 and sigma = 16. New methods were introduced. Has there been a CHANGE in production?

Step 1	H0: mu = 200 | H1: mu != 200 *Keyword: 'has changed' --> use != --> two-tailed test*
Step 2	alpha = 0.01 *Level of significance given in problem*
Step 3	Use Z-distribution since sigma = 16 is KNOWN *Z = (x-bar - mu) / (sigma / sqrt(n))*
Step 4	Reject H0 if |Z| > Z(alpha/2) = Z(0.005) = 2.576 *Two-tailed: rejection regions in both tails*
Step 5	Z_computed = 1.55. Since |1.55| < 2.576, do NOT reject H0. *Conclusion: The production rate has NOT significantly changed from 200 per week.*

Q3 | 10 Marks | 2021-22 Re-exam

Explain the 4 possibilities to make the decision to accept or reject the Null hypothesis. Explain with the help of a table. (Type I error, Type II error, Correct decisions)

The Decision Framework

When we make a decision about H0 using sample evidence, there are only two possible realities (H0 is true OR H0 is false) and two possible decisions (Do not reject H0 OR Reject H0). This gives exactly 4 possible outcomes:

The 4 Possibilities Table

	H0 is TRUE (in reality)	H0 is FALSE (in reality)
Decision: Do NOT reject H0	Correct Decision (True Negative) Probability = 1 - alpha	Type II Error (beta) (False Negative) Probability = beta
Decision: REJECT H0	Type I Error (alpha) (False Positive) Probability = alpha	Correct Decision (True Positive) Probability = 1 - beta = Power

Explanation of All 4 Possibilities

Possibility 1 — Correct Decision (Do not reject / H0 true)

H0 is actually TRUE and we correctly DO NOT REJECT it. This is a True Negative — our test correctly concluded no effect exists. Probability = 1 - alpha.

Possibility 2 — Type I Error (Reject / H0 true)

H0 is actually TRUE but we INCORRECTLY REJECT it. We falsely concluded an effect exists. This is a False Positive. Probability = alpha (the level of significance).

Possibility 3 — Type II Error (Do not reject / H0 false)

H0 is actually FALSE but we INCORRECTLY DO NOT REJECT it. We missed a real effect. This is a False Negative. Probability = beta.

Type II Error = beta Probability of missing a real effect

Possibility 4 — Correct Decision (Reject / H0 false)

H0 is actually FALSE and we correctly REJECT it. This is a True Positive. Our test correctly detected a real effect. Probability = 1 - beta = Power of the test.

Type I Error = alpha Also called the significance level of the test

The Relationship Between Alpha and Beta

There is a fundamental trade-off between Type I and Type II errors. For a fixed sample size, decreasing alpha (making it harder to reject H0) will increase beta (making it easier to miss a real effect). The only way to reduce both simultaneously is to increase the sample size n.

Memory trick: Type I Error (alpha): Rejecting H0 when it is TRUE — 'Crying wolf' (false alarm) Type II Error (beta): Accepting H0 when it is FALSE — 'Missing the wolf' (missed detection) Power of the test = 1 - beta = probability of correctly detecting a real effect

Q4 | 5 Marks | 2023-24 Final & Re-exam | REPEATED

What is Type I and Type II Error? Define with examples and explain the relationship between them.

Type I Error (Alpha)

Type I Error = Rejecting H0 when H0 is ACTUALLY TRUE

Probability of Type I Error = alpha = the level of significance

Type I error is defined as the probability of rejecting the null hypothesis when it is actually true. This is denoted by the Greek letter alpha and is also known as the significance level of the test. When we set alpha = 0.05, we accept a 5% chance of making this error.

Example from notes: A manufacturer sets a rule — accept a steel bar lot if sample mean falls between 9,922 psi and 10,078 psi at the 0.05 significance level. A Type I Error occurs if a GOOD lot (true mean = 10,000 psi) is REJECTED simply due to sampling variation.

Type II Error (Beta)

Type II Error = NOT Rejecting H0 when H0 is ACTUALLY FALSE

Probability of Type II Error = beta

Type II error is defined as the probability of accepting (not rejecting) the null hypothesis when it is actually false. This is denoted by the Greek letter beta. Committing a Type II error means we failed to detect a real effect that was present.

Example from notes: Using the same steel bar rule — a Type II Error occurs if a BAD lot (true mean has shifted to, say, 9,970 psi) is ACCEPTED because the sample mean happened to fall within the acceptance region by chance.

Relationship Between Type I and Type II Errors

	H0 is TRUE (in reality)	H0 is FALSE (in reality)

Decision: Do NOT reject H0	Correct Decision (True Negative) Probability = 1 - alpha	Type II Error (beta) (False Negative) Probability = beta
Decision: REJECT H0	Type I Error (alpha) (False Positive) Probability = alpha	Correct Decision (True Positive) Probability = 1 - beta = Power

The Inverse Relationship: Decreasing alpha (tightening significance level) automatically INCREASES beta — making the test less likely to detect a real effect. Decreasing beta (increasing power) automatically INCREASES alpha — making the test more prone to false alarms. The only way to reduce BOTH errors simultaneously is to INCREASE the sample size n. This is why large samples are preferred in hypothesis testing.

Power of the Test = 1 - beta

Power = probability of correctly rejecting a FALSE H0 = probability of detecting a real effect

Q5 | 5 Marks | 2022-23 Re-exam

How is Hypothesis Testing used in Linear Regression? (F-test, t-test on coefficients, p-values in regression output)

Overview

In linear regression, hypothesis testing is used to determine whether predictors significantly explain the variation in the outcome variable. There are two levels of testing: (1) overall model significance using the F-test, and (2) individual coefficient significance using t-tests.

1. The F-Test — Overall Model Significance

The F-test tests whether the regression model as a whole explains a significant amount of variance in the dependent variable — in other words, whether at least one predictor has a non-zero coefficient.

H0: Beta1 = Beta2 = ... = Bk = 0 (No predictor is useful; model explains nothing)

H1: At least one Beta_i is not equal to 0 (At le

F-statistic formula:

F = (SSR / k) / (SSE / (n - k - 1))

SSR = regression sum of squares SSE = error sum of squares k

Reject H0 if F > F_critical or if p-value (F) < alpha

2. T-Tests on Individual Coefficients

Each regression coefficient Beta_j is tested individually to see whether that specific predictor has a statistically significant relationship with the outcome:

H0: Beta_j = 0 (Predictor Xj has NO effect on Y)

H1: Beta_j != 0 (Predictor Xj DOES affect Y)

t-statistic formula:

t = Beta_hat_j / SE(Beta_hat_j)

SE = standard error of the coefficient estima

Degrees of freedom = n - k - 1 Reject H0 if |t| > t_critical or p-valu

3. P-Values in Regression Output

The p-value is the probability of observing a test statistic as extreme as (or more extreme than) the calculated value, assuming H0 is true. In regression output:

p-value range	Interpretation	Decision
$p < 0.001$	Extremely strong evidence against H0	**Reject H0**
$p < 0.01$	Very strong evidence against H0	**Reject H0**
$p < 0.05$	Strong evidence against H0	**Reject H0**
$p >= 0.05$	Insufficient evidence against H0	**Do NOT reject H0**

Q6 | 5 Marks | 2021-22 Re-exam

Define Bootstrapping and its use in statistics.

What is Bootstrapping?

Bootstrapping is a resampling technique used to estimate statistics on a population by sampling a dataset WITH REPLACEMENT. It allows us to estimate the sampling distribution of almost any statistic using only the original sample data — without making assumptions about the underlying population distribution.

How Bootstrapping Works — Steps

1. **Start with your original sample**
   Take the original dataset of size n. This is your only data.

2. **Draw a bootstrap sample**
   Randomly draw n observations FROM your original sample WITH REPLACEMENT. This means the same observation can appear more than once in a bootstrap sample.

3. **Compute the statistic of interest**
   Calculate whatever statistic you want (mean, median, regression coefficient, etc.) from this bootstrap sample.

4. **Repeat many times**
   Repeat steps 2-3 a large number of times (typically B = 1000 or more). This builds up a bootstrap distribution of your statistic.

5. **Analyse the bootstrap distribution**
   Use the bootstrap distribution to estimate: standard error, confidence intervals, bias of the estimator, and p-values for hypothesis tests.

Uses of Bootstrapping in Statistics

- Estimating confidence intervals when the theoretical distribution is unknown or complex.
- Computing standard errors for estimators where no closed-form formula exists.
- Hypothesis testing without parametric assumptions (non-parametric bootstrap test).
- Model validation — bootstrap samples are used to assess the stability of regression models.

- Estimating the bias of an estimator.
- Works well when the sample size is small and the population distribution is non-normal.

> Key advantage: Bootstrapping makes NO assumptions about the shape of the population distribution. It is especially useful when classical parametric methods (t-tests, Z-tests) cannot be applied because normality or other assumptions are violated.

Q7 | 5 Marks | 2024-25 Final & 2022-23 Re-exam | REPEATED

Explain ANOVA with a suitable example and assumptions. (H0, F-statistic formula, SSB, SSW, decision rule)

What is ANOVA?

ANOVA (Analysis of Variance) is a statistical method used to test whether there are significant differences between the MEANS of THREE OR MORE groups. Instead of running multiple t-tests (which inflates the Type I error rate), ANOVA tests all groups simultaneously using a single F-test.

Why not just use multiple t-tests? If you compare 3 groups with three t-tests at alpha=0.05, the overall chance of making at least one Type I error = 1 - (0.95)^3 = 14.3% — much higher than the intended 5%. ANOVA controls this error rate.

Hypotheses in ANOVA

H0: mu1 = mu2 = mu3 = ... = muk (All group r	H1: At least one group mean is DIFFERENT (Not all means are e

4 Assumptions of ANOVA

Assumption 1 — Independence: All observations must be independent of each other. The sample must be randomly selected from the population.

Assumption 2 — Normality: The data in each group must be approximately normally distributed. ANOVA is robust to mild departures from normality with large samples.

Assumption 3 — Homogeneity of Variance (Homoscedasticity): The variance within each group must be approximately equal across all groups. This is tested with Levene's test or Bartlett's test.

Assumption 4 — Random Sampling: Data must be collected through random sampling to ensure the sample is representative of the population.

The F-Statistic and SS Decomposition

ANOVA decomposes total variation in the data into two parts:

SSB — Sum of Squares BETWEEN groups

Measures variation between group means and the overall mean. This is the variation EXPLAINED by the grouping factor.

SSW — Sum of Squares WITHIN groups

Measures variation within each group — the random variation that CANNOT be explained by the grouping factor.

SSB = Sum[n_i * (x-b

n_i = size of group i; x-bar_i = mean of group i, x-bar_grand = overall mean

SSW = Sum Sum[(x_ij - x-bar_i)^

x_ij = each observation; x-bar_i = mean of its g

SST = SSB + SSW

Total variation = Between-group variation + Within-group variation

F = (SSB / df_between) / (SSW / df_within) = MSB / MSW

df_between = k - 1 | df_within = N - k | k = number of groups | N = total observations

Decision Rule

Reject H0 if: F_calculated > F_critical(k-1, N-k, alpha) A large F means the between-group variation is much larger than the within-group variation — strong evidence that the group means differ. Alternatively: Reject H0 if p-value < alpha

Simple Example

Scenario: A teacher wants to test whether 3 different teaching methods (A, B, C) produce different mean exam scores. H0: mu_A = mu_B = mu_C (all methods give the same average score) H1: At least one method gives a different average score She collects scores from 10 students per method, computes SSB (variation between methods) and SSW (variation within students in the same method), then computes F = MSB/MSW. If F > F_critical at alpha=0.05, she rejects H0 and concludes the methods differ.

Q8 | 5 Marks | 2024-25 Final

Identify which regression assumptions are violated when: (i) scatterplot shows a curve, (ii) residuals show increasing variance, (iii) residuals are non-normal. Explain the consequences of each.

Overview — Regression Assumptions

Linear regression relies on several assumptions about the relationship between variables and the behaviour of the error term (residuals). When these are violated, the OLS estimators may no longer be Best Linear Unbiased Estimators (BLUE) — leading to incorrect inferences.

(i) Scatterplot Shows a Curve — Violation: Non-Linearity

Assumption violated: LINEARITY — the relationship between X and Y must be linear

When the scatterplot of Y vs X shows a curved (non-linear) pattern instead of a straight line, the linearity assumption is violated. Similarly, if a plot of residuals vs fitted values shows a systematic U-shape or curve, non-linearity is present.

Consequences:

- The regression model is misspecified — it is fitting the wrong functional form to the data.
- Coefficient estimates are BIASED — they do not represent the true relationship.
- Predictions will be systematically wrong across the range of X.
- Residuals will show a pattern (not random), making all inference unreliable.

Fix: Apply a transformation (log Y, sqrt Y, X^2) or fit a polynomial/nonlinear model.

(ii) Residuals Show Increasing Variance — Violation: Heteroscedasticity

Assumption violated: HOMOSCEDASTICITY — the variance of errors must be constant (equal) across all levels of X

When residuals show a fan or funnel shape (variance increasing or decreasing with the fitted values), this is called Heteroscedasticity. The error variance is not constant — it depends on X. A residual plot that fans outward is the classic signal.

Consequences:

• OLS estimators remain UNBIASED but are no longer EFFICIENT (not minimum variance).

• Standard errors of coefficients are INCORRECT — leading to wrong t-statistics and p-values.

• Confidence intervals and hypothesis tests become unreliable.

• The model may appear to fit poorly or unevenly across the range of X.

> Detection: Breusch-Pagan test or White's test. Fix: Use Weighted Least Squares (WLS) or transform the dependent variable (e.g. log Y).

(iii) Residuals Are Non-Normal — Violation: Normality of Errors

> Assumption violated: NORMALITY — the error terms must be normally distributed

When the residuals do not follow a normal distribution — shown by a Q-Q plot that deviates from the diagonal, or a histogram of residuals that is skewed — the normality assumption is violated. This matters most in small samples.

Consequences:

• In LARGE samples: by the Central Limit Theorem, violations are less serious — t and F tests remain approximately valid.

• In SMALL samples: t-tests, F-tests, and confidence intervals are INVALID.

• Coefficient estimates themselves (OLS) remain unbiased — normality is only needed for inference.

• Predictions and confidence intervals will be unreliable.

> Detection: Shapiro-Wilk test, Q-Q plot. Fix: Transform Y (log, Box-Cox), remove outliers, or use robust regression methods.

Q9 | 5-10 Marks | 2021-22 Final & Re-exam | REPEATED

Explain the concept of Autocorrelation. Define its causes and effects on regression estimates.

What is Autocorrelation?

Autocorrelation (also called serial correlation) occurs when the ERROR TERMS in a regression model are correlated with each other across observations. In other words, the residual at one time point is related to the residual at a previous time point. This violates the OLS assumption that errors are independent.

OLS Assumption violated: Cov(epsilon_i, epsilon_j) = 0 for all i != j

When this fails, consecutive errors are correlated — positive or negative autocorrelation

Types of Autocorrelation

Positive autocorrelation

A positive residual tends to be followed by another positive residual, and a negative by another negative. The errors cluster together in runs. Most common in economic and time-series data.

DW statistic close to 0 indicates positive autocorrelation

Negative autocorrelation

A positive residual tends to be followed by a negative residual and vice versa. The errors alternate in sign. Less common but occurs in certain economic time series.

DW statistic close to 4 indicates negative autocorrelation

Causes of Autocorrelation

• Omitted variables: An important variable has been left out of the model. Its effect shows up in the residuals, which then follow a pattern.

• Misspecified model: The wrong functional form is used (e.g. a linear model for a non-linear relationship). The misfit creates systematic patterns in residuals.

• Inertia in time-series data: Economic variables like GDP, consumption, and inflation tend to move slowly and carry momentum from one period to the next.

• Data manipulation: Interpolation or smoothing of raw data introduces correlations between adjacent observations.

• Lagged effects: A stimulus in one period affects the outcome in multiple subsequent periods (e.g. a policy change has effects that persist over time).

Effects of Autocorrelation on OLS Estimates

Effect 1: OLS estimators remain UNBIASED — the coefficient estimates are still correct on average.

Effect 2: OLS estimators are NO LONGER EFFICIENT — they do not have minimum variance. Better (more efficient) estimators exist (e.g. GLS — Generalised Least Squares).

Effect 3: Standard errors are UNDERESTIMATED — OLS understates the true uncertainty in the coefficients when positive autocorrelation is present.

Effect 4: t-statistics are INFLATED — because standard errors are too small, the t-values appear larger than they really are.

Effect 5: Hypothesis tests are INVALID — p-values are wrong, leading to incorrectly rejecting H0 and concluding predictors are significant when they are not.

Effect 6: R-squared may be OVERESTIMATED — the model appears to fit better than it actually does.

Detection — The Durbin-Watson Test

The Durbin-Watson (DW) statistic is the standard test for autocorrelation in regression residuals:

DW = Sum[(e_t - e_(t-1))^2] / Sum[e_t^2]

Where e_t is the residual at time t

DW Value	Interpretation
Close to 0	**Strong POSITIVE autocorrelation**
Close to 2	**NO autocorrelation (desirable)**
Close to 4	**Strong NEGATIVE autocorrelation**

Fix for autocorrelation: 1. Add omitted variables to the model. 2. Use Generalised Least Squares (GLS) or Cochrane-Orcutt procedure. 3. Include lagged variables in the model. 4. Use HAC (Heteroscedasticity and Autocorrelation Consistent) standard errors (also called Newey-West standard errors).